A

TA Rank SEQ

TB Rank SEQ

ICL11 ICL10 ICL9 SCL9 ICL8 SCL8 ICL7 SCL7 ICL6 SCL6 ICL5 ICL4 ICL3 ICL2 ICL1

0 100

Time (ms)

1 0

Normalized magnitude

2D projection

SCL

ICL

10 mm

σ = 6 mm

TA Spatial TRAJ.

TB Spatial TRAJ.

ranks

1 late

0.5

0 early

y (mm)

20 0 −20

shared TA axis (mm)

B

Rank SEQ

Spatial TRAJ.

SEQ null

TRAJ. null

Patients

E14

E10

***

Relative density

1 0

0 30 60 90

Held-out error (°)

C

E10

E14

90° 20% 10% 180° 0° 270°

Event directions

Spatial TRAJ.

D

E1 E5 Y9 Y14

rank

late

early

TA TRAJ.

TB TRAJ.

y (mm)

10 0 −10 20

−10 0 10 −20 0 20

Shared TA axis (mm)

E

negative: **

15/18 patients

E18 Y16 E6 Y6 Y15 E15 Y5 E10 Y14 E5 E3 E2 E12 Y9 Y3 E9 E1 E20

***

−1.0 −0.5 0.0 0.5 1.0

Cohort median r vs spatial null